

Hardware Schematic & Wiring Specification

Project: AI stick: Predictive Mobility & Fall Detection System

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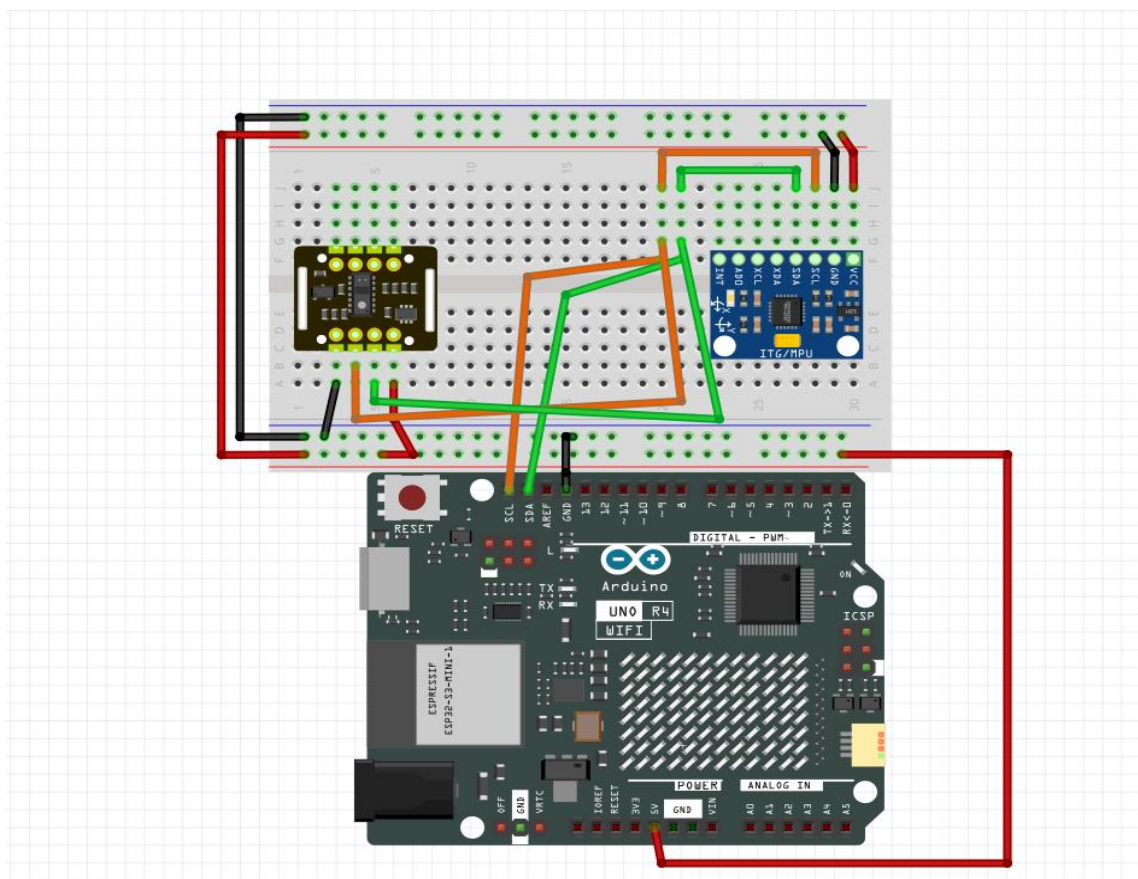
1. System Overview

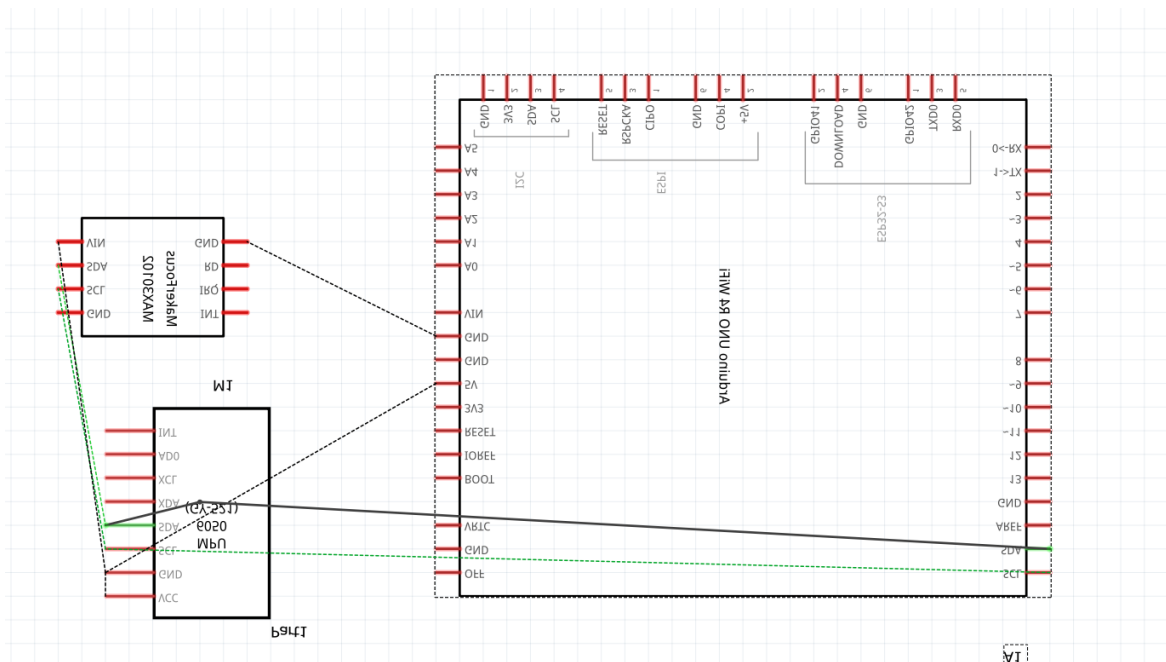
This document details the electrical wiring and I2C communication pathways for the AI stick hardware node. The system integrates a central microcontroller with dual biomedical and biomechanical sensors to capture real-time human telemetry.

2. Component Roster

- **Microcontroller:** Arduino UNO Q (Qualcomm Snapdragon architecture)
- **Biomechanical Sensor (IMU):** MPU-6050 (6-DoF Accelerometer & Gyroscope)
- **Biomedical Sensor (Pulse Oximetry):** MAX30102 (High-Sensitivity SpO2 & Heart Rate)
- **Infrastructure:** Standard breadboard and jumper wiring

3. Wiring Diagrams





Important Engineering Note Regarding CAD Diagrams: The Fritzing diagrams above utilize the symbol for the Arduino UNO R4 WiFi as a visual placeholder. The physical deployment of the AI stick utilizes the **Arduino UNO Q**. Because the UNO Q shares the standard Arduino R3/R4 physical pinout footprint, the electrical pathways, power distribution, and I2C bus connections depicted in these diagrams remain 100% accurate to the physical build.

4. Pin Connection Mapping

The system utilizes the I2C (Inter-Integrated Circuit) protocol to allow both sensors to communicate with the microcontroller over the same two data lines.

Arduino UNO Q Pin	MPU-6050 Pin	MAX30102 Pin	Function Description
5V	VCC	VIN	Main logic power supply
GND	GND	GND	Common ground reference
SCL (or Pin A5)	SCL	SCL	I2C Serial Clock Line
SDA (or Pin A4)	SDA	SDA	I2C Serial Data Line

5. Power and Communication Architecture

- **I2C Bus Sharing:** Both the MPU-6050 and the MAX30102 are connected in parallel to the Arduino's SCL and SDA pins. The Arduino distinguishes between the two sensors in the C++ software using their unique, hardcoded I2C hexadecimal addresses.
- **Power Distribution:** The 5V pin from the Arduino UNO Q routes to the positive power rail of the breadboard, which then distributes power to the VCC/VIN pins of both breakout boards. Both sensors share a common ground loop back to the microcontroller to prevent voltage floating.